



PTT PHENOL COMPANY
LIMITED

PROCESS SPECIFICATION

ANALYZER PROCESS DATA SHEET

Docu. No. : 14780-8120-PS-0003
Project No. : 120117
Rev. No. : Z1
Date : MAY 10, 2016
Page : 1 / 12

TITLE :

ANALYZER PROCESS DATA SHEET

(CDN SECTION)

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DOCUMENT NO. : 14780-8120-PS-0003
PROJECT NAME : PTT Phenol Train II Project
PROJECT NO. : 120117
OWNER : PTT Phenol Company Limited (PPCL)

Z1	MAY 10, 2016	AS BUILT	S.W.Park	J.S.Chung	-	S.W.Lee
F1	JUL 17, 2014	FINAL ISSUE	S.W.Park	J.S.Chung	-	S.W.Lee
A1	MAY 13, 2014	APPROVED FOR DESIGN	S.W.Park	J.S.Chung	-	S.W.Lee
O2	DEC 06, 2013	COMMENT ISSUE	S.W.Park	J.S.Chung	-	S.W.Lee
O1	OCT 7, 2013	COMMENT ISSUE	S.W.Park	J.S.Chung	-	S.W.Lee
REV. NO.	DATE	DESCRIPTION	PREP'N	REVIEW	QA REVIEW	APPROVAL
REV. NO.	PREPARATION DEPT.	PREPARATION	REVIEW	QA REVIEW	APPROVAL	
Z1	PROCESS	S.W.Park	J.S.Chung	-	S.W.Lee	
		MAY 10, 2016	MAY 10, 2016	-	MAY 10, 2016	

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ANALYZERS

PROJECT SPECIFICATION

963766

SHEET

3/12

REV	DATE	BY	APVD	REV	DATE	BY	APVD

Concentration, Decomposition and Neutralization Section

Item Number	Service	Sheet
Oxygen Analyzers		
AT-23-1001	Concentration Vacuum Producing Equipment Vent	4
Density Instruments		
AT/AY-23-0801	Preflash and Flash Columns Overhead Liquid	7
AT/AY-23-0901	Flash Column Bottoms	8
Sulfuric Acid Analyzer		
AT/AY-23-1701	Decomposer Circulating Liquid	10
AT/AY-23-1901	Crude Product to Fractionation Feed Tanks at Fractionation Section	11

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OXYGEN ANALYZERS

PROJECT SPECIFICATION

963766 - 630

SHEET

4/12

REV	DATE	BY	APVD	REV	DATE	BY	APVD
0	25JUN12	MVD	MW				

Item Quantity

1 1

Description

Oxygen analyzer transmitter complete with sampling system including sample gas / calibration gas selector panel for field location. All metal parts in contact with the sample to be 316 stainless steel. Oxygen analyzer shall measure percent oxygen in a vent gas stream.

Tag Number: AT-23-1001

Service: Concentration Vacuum Producing Equipment Vent

Analyzer Unit

Type: Paramagnetic Oxygen Analyzer Transmitter

Range: 0 - 10%

Accuracy: $\pm 2\%$ of range

Transmitter Output: 4 - 20 mA

Sample Conditions

Sample: Vapor

Pressure: 0.07 kg/cm²(g)

Temperature: 20°C

Specific Gravity: 0.981

Approximate Sample Composition

Component	Mole %
Oxygen	4.97
Nitrogen	92.51
Water	2.17
Cumene	0.31
Acetone	Trace
Formic	Trace

Sample System

Sample system shall be complete with pressure reducing regulator, filter moisture separator, and sample pump (if required). See Sketch "A".

Other specifications
referenced in this
specification:

UOP Standard Specification:
UOP Standard Drawing:
UOP Project Specification: 601, this project

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OXYGEN ANALYZERS

PROJECT SPECIFICATION

963766 - 630

SHEET

5/12

REV	DATE	BY	APVD	REV	DATE	BY	APVD
0	25JUN12	MVD	MW				

Item

1 (cont'd)

Description

Utilities Available

Electrical Power: 220 Volts, 50 Hz.
Cooling Water: 33°C at 3.9 kg/cm²(g)

Site Conditions

Ambient Temp. Range: 15 to 40°C

Notes:

1. Vendor to furnish calibration sample specifications, complete wiring, component and installation drawings along with instruction manuals.
2. Vendor shall state degree of weather protection required. Shelter and temperature control equipment shall be provided by contractor.

Supplier (or equal):

Teledyne Analytical Instruments
16830 Chestnut Street
City of Industry, California 91748-1020
USA
www.teledyne-ai.com

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OXYGEN ANALYZERS

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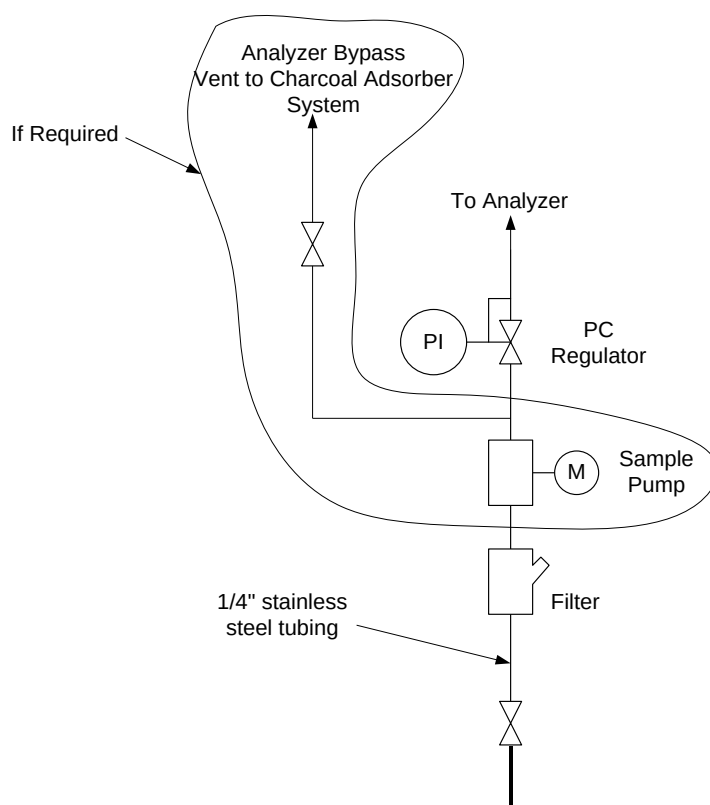
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SHEET

6/12

REV	DATE	BY	APVD	REV	DATE	BY	APVD
0	25JUN12	MVD	MW				

Sketch " A "



Notes:

1. Sample pump per vendor recommendation.
2. Provide weather protection as required.
3. Sample line shall be free draining from analyzer to process line.
4. Deleted.
5. Vent gas shall be discharged to Charcoal Adsorber System.

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DENSITY INSTRUMENTS

PROJECT SPECIFICATION

963766 - 654

SHEET

7/12

REV	DATE	BY	APVD	REV	DATE	BY	APVD
0	25JUN12	MVD	MW				

Item	Quantity	Description				
1	1	Provide a density measurement system for each of the services listed below.				
		<table><tr><th>Tag Number</th><th>Service</th></tr><tr><td>AT/AY-23-0801</td><td>Preflash and Flash Columns Overhead Liquid</td></tr></table>	Tag Number	Service	AT/AY-23-0801	Preflash and Flash Columns Overhead Liquid
Tag Number	Service					
AT/AY-23-0801	Preflash and Flash Columns Overhead Liquid					
1		Density measurement transmitter with sensing spool and amplifier for liquid density measurement. Density meter body shall be 316 stainless steel with 1", Class 150 (minimum) RF flanged connections. Body shall have built-in platinum resistance temperature detector for spool characteristic compensation. Accuracy of measurement shall be ±0.0001 g/cm ³ . Sensing spool material shall be 316 stainless steel.				
		Operating Conditions				
		Pressure (normal): 7.394 kg/cm ² (g)				
		Temperature (variation): 39°C normal (30 - 80 °C)				
		Specific Gravity (normal): 0.848				
		Required Operating Density Span: 700 – 1000 kg/m ³				
		Sample is Cumene with Cumene Hydroperoxide (CHP).				
		Fluid Phase Liquid				
1		Density converter for CHP concentration measurement for use with density transmitter described above. Converter output shall have a calibrated range of 0 - 10 wt% CHP. The converter shall use a look up table using data provided in Table 1 to interpolate the concentration of CHP for the temperature variation and calibration range specified. The converter will be located in the control center. Converter output for the CHP concentration shall be 4 - 20 mA. The converter shall have a second 4 - 20 mA output that corresponds directly to the Required Operating Density Span above. (contractor to confirm outputs). Power supply is 220 VAC 50 Hz.				
1		Spare density sensing spool with calibration data for use with transmitter. Calibration characteristics to be similar to those for spool supplied with transmitter.				
1		Instruction manuals for density transmitter, converter, etc. with recommended spare parts list.				
Required		Shielded signal cable for sensor-to-converter transmission. Cable type and maximum length to be specified by density monitor vendor. Contractor to purchase specified cables in lengths required.				

Supplier (or equal):
Thermo Fisher Scientific
Sarasota
1410 Gillingham Lane
Sugar Land, Texas 77478 USA
www.thermo.com

Other specifications referenced in this specification:	UOP Standard Specification:
	UOP Standard Drawing:
	UOP Project Specification: 601, this project

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DENSITY INSTRUMENTS

PROJECT SPECIFICATION

963766 - 654

SHEET

8/12

REV	DATE	BY	APVD	REV	DATE	BY	APVD
0	25JUN12	MVD	MW				

Item	Quantity	Description				
2	1	Provide a density measurement system for each of the services listed below.				
		<table><tr><th>Tag Number</th><th>Service</th></tr><tr><td>AT/AY-23-0901</td><td>Flash Column Bottoms</td></tr></table>	Tag Number	Service	AT/AY-23-0901	Flash Column Bottoms
Tag Number	Service					
AT/AY-23-0901	Flash Column Bottoms					
	1	Density measurement transmitter with sensing spool and amplifier for liquid density measurement. Density meter body shall be 316 stainless steel with 1", Class 150 (minimum) RF flanged connections. Body shall have built-in platinum resistance temperature detector for spool characteristic compensation. Accuracy of measurement shall be ±0.0001 g/cm ³ . Sensing spool material shall be 316 stainless steel.				
		Operating Conditions				
		Pressure (normal): 3.384 kg/cm ² (g)				
		Temperature (variation): 60°C normal (40 - 90°C)				
		Specific Gravity (normal): 0.972				
		Required Operating Density Span: 800 – 1100 kg/m ³				
		Sample is Cumene with Cumene Hydroperoxide (CHP)				
		Fluid Phase: Liquid				
	1	Density converter for CHP concentration measurement for use with density transmitter described above. Converter output shall have a calibrated range of 50 - 90 wt% CHP. The converter shall use a look up table using data provided in Table 1 to interpolate the concentration of CHP for the temperature variation and calibration range specified. The converter will be located in the control center. Converter output for CHP concentration shall be 4 - 20 mA. The converter shall have a second 4 - 20 mA output that corresponds directly to the Required Operating Density Span above. (contractor to confirm outputs) Power supply is 220 VAC 50 Hz.				
	1	Spare density sensing spool with calibration data for use with transmitter. Calibration characteristics to be similar to those for spool supplied with transmitter.				
	1	Instruction manuals for density transmitter, converter, etc. with recommended spare parts list.				
Required		Shielded signal cable for sensor-to-converter transmission. Cable type and maximum length to be specified by density monitor vendor. Contractor to purchase specified cables in lengths required.				

Supplier (or equal):

Thermo Fisher Scientific
Sarasota
1410 Gillingham Lane
Sugar Land, Texas 77478 USA
www.thermo.com

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DENSITY INSTRUMENTS

PROJECT SPECIFICATION

963766 - 654

SHEET

9/12

REV	DATE	BY	APVD	REV	DATE	BY	APVD
0	25JUN12	MVD	MW				

Table 1

DENSITIES OF SOLUTIONS - CHP IN CUMENE (g/cm ³)										
	wt% Cumene Hydroperoxide									
°C	0%	10%	20%	30%	40%	50%	60%	70%	80%	90%
20	0.862	0.878	0.896	0.915	0.937	0.959	0.983	1.006	1.030	1.055
25	0.858	0.874	0.892	0.911	0.933	0.955	0.979	1.002	1.026	1.051
30	0.854	0.870	0.888	0.906	0.928	0.950	0.975	0.997	1.022	1.046
35	0.849	0.865	0.883	0.902	0.924	0.946	0.970	0.993	1.017	1.042
40	0.845	0.861	0.879	0.898	0.920	0.942	0.966	0.989	1.013	1.038
45	0.841	0.856	0.875	0.894	0.916	0.938	0.961	0.985	1.009	1.034
50	0.837	0.852	0.870	0.890	0.912	0.934	0.957	0.981	1.005	1.030
55	0.832	0.848	0.866	0.886	0.908	0.930	0.953	0.977	1.001	1.026
60	0.828	0.844	0.862	0.882	0.904	0.925	0.948	0.972	0.996	1.021
65	0.824	0.840	0.858	0.878	0.900	0.921	0.944	0.968	0.992	1.017
70	0.820	0.835	0.854	0.874	0.895	0.917	0.940	0.964	0.988	1.013
75	0.816	0.831	0.850	0.870	0.891	0.913	0.936	0.960	0.984	1.009
80	0.811	0.827	0.845	0.865	0.887	0.908	0.932	0.956	0.980	1.005
85	0.807	0.822	0.841	0.861	0.882	0.904	0.927	0.951	0.976	1.000
90	0.802	0.818	0.837	0.856	0.878	0.900	0.923	0.947	0.971	0.996
95	0.798	0.814	0.833	0.853	0.874	0.896	~	~	~	~
100	0.794	0.810	0.828	0.848	0.870	0.892	~	~	~	~

Note:

Data shown represents a pure two-component system. Actual CHP concentration may be different than the indicated value of the on-stream monitor and is dependant upon unit operations. Lab analysis for CHP concentration should be routinely submitted for verification of the on-stream measurement.

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SULFURIC ACID ANALYZERS

PROJECT SPECIFICATION

963766 - 656

SHEET

10/12

REV	DATE	BY	APVD	REV	DATE	BY	APVD
0	25JUN12	MVD	MW				

Item Quantity

Description

1 1

Provide a colorimetric titration type sulfuric acid analyzer with sample conditioner. Vendor shall determine suitable analytical method for the composition shown below. The analyzer shall be designed for acid-base titration analysis using a known caustic solution titrated to a fixed pH.

Tag Number: AT, AY-23-1701

Service: Decomposer Circulating Liquid

Analyzer

Scale Range: 0 to 350 wt. ppm H₂SO₄

The analyzer section will be located outdoors. The vendor shall furnish the analyzer section completely assembled for installation in a temperature controlled housing (as required). The vendor shall provide certified drawings of the analyzer including an illustration of the analyzer mounting brackets. The temperature controlled housing shall be provided by the vendor or by the Contractor based on the analyzer vendor's recommendation. The vendor shall state the range of temperature variation permissible in the housing to reduce temperature induced measurement error to less than 0.5 % of full scale.

The Contractor shall provide heating and/or cooling, as required, to maintain the housing temperature within the recommended range for an ambient temperature range of 15 to 40°C.

The wet section (fluidics) shall consist of a sample/reaction cell assembly containing a sensor, stirrer assembly, reagent, calibrant, and titrant pumps. Reaction cup material shall not be PTFE. Valve actuators, motors, and electrical connectors are back panel mounted behind the fluidic section. Sample flow is controlled by the inlet valve and the air vent valve.

Sample Conditions

Supply: 3.899 kg/cm² (g)
Return: 2.915 kg/cm² (g)
Sp. Gr.: 0.898 at 79°C
H₂SO₄ Content: 40 wt. ppm (300 wt. ppm during start-up)
Fluid Phase: Liquid

Approximate Sample Composition

Component	Wt %
Phenol	48.94
Acetone	30.25
Cumene	12.54
Dicumylperoxide	3.91
CHP	1.87
Water	0.93

Other specifications
referenced in this
specification:

UOP Standard Specification:
UOP Standard Drawing: 6-152
UOP Project Specification: 601, this project

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SULFURIC ACID ANALYZERS

PROJECT SPECIFICATION

963766 - 656

SHEET

11/12

REV	DATE	BY	APVD	REV	DATE	BY	APVD
0	25JUN12	MVD	MW				

Item Quantity

Description

2 1

Provide a colorimetric titration type sulfuric acid analyzer with sample conditioner. Vendor shall determine suitable analytical method for the composition shown below. The analyzer shall be designed for acid-base titration analysis using a known caustic solution titrated to a fixed pH.

Tag Number: AT, AY-23-1901

Service: Crude Product to Fractionation Feed Tanks at Fractionation Section

Analyzer

Scale Range: 20 to 60 wt. ppm H₂SO₄

The analyzer section will be located outdoors. The vendor shall furnish the analyzer section completely assembled for installation in a temperature controlled housing (as required). The vendor shall provide certified drawings of the analyzer including an illustration of the analyzer mounting brackets. The temperature controlled housing shall be provided by the vendor or by the Contractor based on the analyzer vendor's recommendation. The vendor shall state the range of temperature variation permissible in the housing to reduce temperature induced measurement error to less than 0.5 % of full scale.

The Contractor shall provide heating and/or cooling, as required, to maintain the housing temperature within the recommended range for an ambient temperature range of 15 to 40°C.

The wet section (fluidics) shall consist of a sample/reaction cell assembly containing a sensor, stirrer assembly, reagent, calibrant, and titrant pumps. Reaction cup material shall not be PTFE. Valve actuators, motors, and electrical connectors are back panel mounted behind the fluidic section. Sample flow is controlled by the inlet valve and the air vent valve.

Sample Conditions

Supply: 2.38 kg/cm² (g)

Return: 1.17 kg/cm² (g)

Sp. Gr.: 0.930 at 43°C

H₂SO₄ Content: 38 wt. ppm

Fluid Phase: Liquid

Approximate Sample Composition

Component	Wt %
Phenol	51.31
Acetone	30.76
Cumene	13.48
Alpha-Methylstyrene	2.05
Water	1.11
Cumyl Phenols	0.48

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SULFURIC ACID ANALYZERS

PROJECT SPECIFICATION

963766 - 656

SHEET

12/12

REV	DATE	BY	APVD	REV	DATE	BY	APVD
0	25JUN12	MVD	MW				

Item

1 - 2 (cont'd)

Description

Sample System

Vendor shall supply all necessary equipment for sample circulation, filtration, flow regulation, including rotameters for fast loop circulation and sample. Contractor shall supply pressure reducing regulator and flow regulation equipment for nitrogen and air.

The process fast loop circulation piping shall be arranged to require a maximum length of 12 inches (300 mm) from the sample source (sample tap) to the sample conditioning system inlet. See UOP Standard Drawing 6-152 for similar installation.

Contractor shall supply piping for draining spent sample, reagent and flush material to both Acid Aromatics Sump and Phenolic Waste Water Sump. Vendor shall furnish complete installation drawings, utility requirements, operating instruction and drawings, calibration sample specifications and calibration procedures, and trouble shooting instructions. Vendor shall furnish a one year supply of sensing tape, reagent and calibration fluids, and recommended spare parts.

Programmer

The control unit shall be mounted integral with the analyzer in an enclosure suitable for both the environment and the area electrical classification. The microprocessor-based field programmable control unit shall be complete with power supply, integral output indication and isolated output signal. The control unit output signal shall be sent to a distributed control system via 4 to 20 mA current loop.

Calibration shall be automatic and fully programmable through menu driven programming. The analysis frequency shall be programmable with initial setting of 15 minutes. A diagnostics function key shall be provided.

Service Engineer

The vendor shall furnish a competent service engineer to review installation and supervise calibration and initial startup. Proposal shall state terms of this service.

Utilities Available

Electrical Power:	220 Volts, 50 Hz.
Air:	7.0 kg/cm2(g) at 38°C
Process Water:	5.0 kg/cm2(g) at 50°C
Nitrogen:	7.0 kg/cm2(g) at 38°C
Steam:	135°C at 1.5 kg/cm2(g)

Supplier (or equal):

Galvanic Applied Sciences
41 Wellman Street
Lowell, Massachusetts 01851 USA
www.galvenic.com